

Week 1 introduction solution

1. What is a decision tree?

A decision tree shows how a conclusion is taken based on choices. A decision tree ask a question based on answers from previous questions. After answering all the questions a conclusion is taken.

2. What is rule-based?

If-then-else statements

3. What is greedy approach in decision tree?

Greedy approach is when you ask a question, and choose the next question which is the most informative given the first answer

4. What is overfitting? How can we overfit? How can we avoid overfitting in decision trees?

Overfitting is when the model fits the training data too well. This can happen if you ask too many questions.

Avoid overfitting by stopping the questions earlier (before reaching leave nodes).

5. What is supervised learning?

Supervised learning is when we have the input set and desired output in training. We know the labels to the input. We have examples on the problem and answers. Requires someone to label the data, which can be labor consuming. The requirement to label the data limits the number of applicable problems.

6. What is unsupervised learning?

Unsupervised learning is when we have the input, but we do not know the desired output. It is normal to group the input into groups based on similarities to label them. How many groups to make is chosen by the user.

7. What is reinforcement learning?

Reinforced learning is when we have an entity, agent, in an environment. We only tell the agent when an action is bad or good, based a rewards. It has to figure out how to change its behavior to get more rewards